

JavaScript Comparisons

[Back to JS page](#)

Table of Contents

- [JavaScript Comparisons](#)
 - [Comparison Operators](#)
 - [Example 1](#)
 - [JavaScript String Comparison](#)
 - [Example 2](#)
 - [Comparing Different Types](#)
 - [Example 3](#)
 - [Document](#)
 - [Reference](#)

Comparison Operators

Comparison operators are used to **compare two values** and always return `true` or `false`.

Given that `x = 5`, the table below explains the comparison operators:

Operator	Description	Comparing	Returns
<code>==</code>	equal to	<code>x == 8</code>	false
<code>==</code>	equal to	<code>x == 5</code>	true
<code>==</code>	equal to	<code>x == "5"</code>	true
<code>===</code>	equal value and type	<code>x === 5</code>	true
<code>===</code>	equal value and type	<code>x === "5"</code>	false
<code>!=</code>	not equal	<code>x != 8</code>	true
<code>!==</code>	not equal value or type	<code>x !== 5</code>	false
<code>!==</code>	not equal value or type	<code>x !== "5"</code>	true
<code>></code>	greater than	<code>x > 8</code>	false
<code><</code>	less than	<code>x < 8</code>	true
<code>>=</code>	greater than or equal to	<code>x >= 8</code>	false
<code><=</code>	less than or equal to	<code>x <= 8</code>	true

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Comparisons</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Comparison Operators</h4>
<p id="demo"></p>

<script>
let x = 5;
document.getElementById("demo").innerHTML =
  "x = " + x + "<br><br>" +
  "x == 8 : " + (x == 8) + "<br>" +
  "x == 5 : " + (x == 5) + "<br>" +
  "x == '5' : " + (x == "5") + "<br>" +
  "x === 5 : " + (x === 5) + "<br>" +
  "x === '5' : " + (x === "5") + "<br>" +
  "x != 8 : " + (x != 8) + "<br>" +
  "x > 8 : " + (x > 8) + "<br>" +
  "x < 8 : " + (x < 8) + "<br>" +
  "x >= 8 : " + (x >= 8) + "<br>" +
  "x <= 8 : " + (x <= 8);
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

JavaScript String Comparison

All the comparison operators can also be used on strings. Note that strings are compared alphabetically:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Comparisons</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>String Comparison</h4>
<p id="demo"></p>

<script>
let text1 = "A";
let text2 = "B";
let result = text1 < text2;
document.getElementById("demo").innerHTML =
  "A < B : " + result + "<br><br>" +
  "\"Apple\" < \"Banana\" : " + ("Apple" < "Banana") + "<br>" +
  "\"John\" > \"Jane\" : " + ("John" > "Jane");
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

Comparing Different Types

Comparing data of different types may give unexpected results.

When comparing a string with a number, JavaScript will convert the string to a number when doing the comparison.

An empty string converts to 0. A non-numeric string converts to `NaN` which is always `false` :

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Comparisons</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Comparing Different Types</h4>
<p id="demo"></p>

<script>
document.getElementById("demo").innerHTML =
  "2 < 12 : " + (2 < 12) + "<br>" +
  "\"2\" < \"12\" : " + ("2" < "12") + "<br>" +
  "\"2\" < 12 : " + ("2" < 12) + "<br>" +
  "\"\" == 0 : " + ("\" == 0) + "<br>" +
  "\"\" === 0 : " + ("\" === 0);
</script>

</body>
</html>
```

Example 3

Result [View Example](#)

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript Comparisons](#)